THE CLEBSCH CUBIC PROGRAM — IV

# Minimum-word reconstruction of PG$(2,13)$ from a binary conic code

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**Abstract**

Let $\mathcal{C}$ be a nonsingular conic in PG$(2,13)$, and form the binary incidence code on its 78 internal points using the 78 passant lines. We prove that the code has parameters $[78,36,12]_2$ and exactly 364 minimum words. Their weighted pair concurrences alone reconstruct the passant incidence matrix, the code, and the six-class elliptic association scheme. The resulting group action then reconstructs all points and lines of PG$(2,13)$, the distinguished conic, and its polarity; no coordinates or triple concurrence are required. Equivalently, the weighted 2-section of the minimum-support hypergraph is a complete invariant of this marked conic-plane presentation. The four minimum-word families are one octahedral family and three chord-indexed punctured-conic families, and each spans the code. Structurally, the code is 12-dimensional over a canonical operator field $\mathbb{F}_8$; this hidden scalar action explains why every minimum-word family spans. Exact positive-semidefinite and line-moment certificates exclude weights eight and ten, replacing the corresponding subset and syndrome searches.

**Keywords.** binary incidence code; finite conic; passant line; minimum word; association scheme; reconstruction.

## 1 Introduction and statement

Incidence codes usually remember less than the structures that define them. Row operations erase the chosen parity checks, and minimum distance alone does not explain the geometry of the first codewords. The code studied here behaves in the opposite way: the weighted pairs inside its minimum supports recover not only the parity checks but the full projective plane in which they arose. This is the fourth paper in a program on code-theoretic reconstruction around the Clebsch cubic. The first paper begins from the conic code over $\mathbb{F}_{11}$; the present $q=13$ theorem is logically independent of the earlier papers.

Fix a nonsingular conic $\mathcal{C} \subset$ PG$(2,13)$. A line is *passant* when it is disjoint from $\mathcal{C}(\mathbb{F}_{13})$. Conic polarity identifies the 78 internal points with the 78 passant lines. Let $M$ be their binary incidence matrix and put

$$K = \ker_{\mathbb{F}_2} M \subseteq \mathbb{F}_2^{78}.$$

The coordinate set is the set of internal points. Automorphisms below are coordinate permutations preserving the indicated code or minimum-support system.

Let

$$\mathcal{H} = \{\mathrm{supp}(c) : c \in K,\ \mathrm{wt}(c) = 12\}$$

be the minimum-support hypergraph on the unlabeled coordinate set of $K$.

**Theorem 1.1** (Minimum words recover the conic plane)**.** *The code $K$ has parameters $[78, 36, 12]_2$ and exactly 364 minimum words. Under $\mathrm{PGL}(2, 13)$, those words form four orbits of size 91. One is an octahedral homogeneous space with stabilizer $S_4$; the other three are chord-indexed punctured conics with stabilizer $D_{24}$. Every orbit spans $K$.*

*Let $c(P, Q)$ be the number of minimum supports containing $P, Q$. The neighborhoods defined by $c(P, Q) = 8$ are exactly the 78 rows of $M$. The other five elliptic relations are recovered from the remaining pair weights and one common-neighbor count. Moreover, if $P_{PQ} = c(P, Q) \bmod 2$ off the diagonal and $P_{PP} = 0$, then $\operatorname{im} P = K$. Thus the weighted 2-section of $\mathcal{H}$ reconstructs the incidence matrix, code, and elliptic scheme.*

*From the reconstructed group, its fourteen Sylow-13 subgroups and 169 involutions recover all 183 points and lines of $\mathrm{PG}(2, 13)$, the distinguished conic, and its polarity. Consequently weighted-pair reconstruction has exact arity two, and*

$$\mathrm{Aut}(K) = \mathrm{Aut}(\mathcal{H}) = \mathrm{PGL}(2, 13)$$

*in its symmetric-square action on the 78 internal points.*

Reconstruction is intrinsic, not coordinatized. It recovers the marked plane and polarity up to isomorphism, but it does not choose an ordered projective frame, a field labeling, or an equation for the conic. Such a choice is necessarily noncanonical: the 2184 ordered triples of distinct conic points form a torsor for $\mathrm{PGL}(2, 13)$.

The proof is structural except at small exact endpoints. A rank-28 positive-semidefinite matrix excludes weight eight. A global line moment reduces weight ten to two shapes, which are eliminated on four line- stabilizer orbits and thirty-three point-stabilizer orbits. The minimum layer itself is exhausted at one point, the octahedral construction has one bounded matching-profile check, and the recovered group/plane dictionary is independently replayed in the complete finite model.

The code belongs to the fixed-conic LDPC family introduced by Droms–Mellinger–Meyer [2]. Their general bounds, as recorded in the later comparison table of Ma–Liu–Tian [5, Table 1], specialize here to $8 \leq d \leq 12$. Madison and Wu proved the general nullity formula $(q-1)^2/4$ [4, Corollary 6.3]. Hollmann and Xiang constructed the elliptic association scheme afforded by the same conic action [3, Sections 3–5]. The identification of involutions with off-conic points, with involution axis equal to polar line, is classical and is stated explicitly by Tranchida [6, Section 2]. We determine the upper endpoint of the distance interval at $q = 13$, classify its minimum words, and prove that their weighted pair data recovers these established ambient structures.

# 2 Distance twelve

We use the model $\mathcal{C} : XZ - Y^2 = 0$. Every row and column of $M$ has weight seven, and conic polarity identifies the two index sets. The Madison–Wu formula gives $\dim K = 36$; direct binary elimination is an independent check of rank 42.

Summing all row equations gives the all-one functional, so every word of $K$ has even weight. If a nonzero word contains an internal point $P$, each of the seven passants through $P$ contains a second support point. Those seven points are distinct. Thus every nonzero word has weight at least eight.

## 2.1 The tangent obstruction in weight eight

Suppose that a word has weight eight and fix a support point $P$. Each of the seven passants through $P$ must contain another support point. Since only seven points remain, each passant contains exactly one of them. Applying the same argument at every support point shows that every support join is passant and that no three support points are collinear. Thus the support is an eight-arc, and every passant pencil at a support point is saturated. The seven remaining lines through each support point are therefore both the combinatorial tangents to the arc and the conic secants through that point.

For each secant line choose a nonzero linear equation. For an internal point $P$, put

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple define

$$h(P, Q, R) = \frac{T_P(Q) T_Q(R) T_R(P)}{T_P(R) T_R(Q) T_Q(P)}.$$

Every displayed evaluation is nonzero because the three joins are passant. Rescaling a line equation multiplies one numerator and one denominator factor by the same scalar, so $h$ is well defined. The saturated-pencil identification above makes the factors of $T_P$ precisely the combinatorial tangents at $P$. Segre's lemma of tangents in the coordinate-free form of Ball–Lavrauw [1, Lemma 27] gives $h(P, Q, R) = 1$ for every triple in the hypothetical support. Indeed, its hypotheses apply to this eight-arc over the odd field $\mathbb{F}_{13}$, and with the cyclic order $P, Q, R$ its tangent-product quotient is exactly the displayed $h$ (not its negative; reversing the cycle only inverts it). Fix $P = (1 : 0 : 2)$. Its other seven points would form a clique in the graph on the 42 passant-join neighbors of $P$, where $Q$ and $R$ are adjacent when $QR$ is passant and $h(P, Q, R) = 1$.

The projective map

$$\gamma(x : y : z) = (x + 6y + 9z : 7x + 9y + 3z : 10x + y + z)$$

fixes $P$ and has order 14 on its passant-join neighbors. With indices in $\mathbf{Z}/14$, put

$$A_i = \gamma^i(1 : 1 : 7), \qquad B_i = \gamma^i(1 : 1 : 6), \qquad C_i = \gamma^i(1 : 2 : 10).$$

These three orbits are disjoint and contain all 42 neighbors. Direct substitution in the product above gives

| orbit pair | $j - i \pmod{14}$ |
|---|---|
| $AA$ | $4, 6, 8, 10$ |
| $AB$ | $6, 7, 11, 12$ |
| $AC$ | $1, 3$ |
| $BB$ | $6, 8$ |
| $BC$ | $3, 5, 6, 8, 9, 11$ |
| $CC$ | $2, 4, 10, 12.$ |

Let $C$ be the symmetric 42-square matrix in the displayed vertex order, written in 14-square circulant blocks. Its six upper-block first rows are

$$\begin{array}{r|rrrrrrrrrrrrrr}
00 & 40 & 9 & 8 & -8 & -10 & 0 & -10 & -2 & -10 & 0 & -10 & -8 & 8 & 9 \\
01 & 7 & 6 & 6 & 6 & 7 & -1 & -10 & -10 & -8 & 0 & -8 & -10 & -10 & -1 \\
02 & 0 & -10 & 2 & -10 & 0 & 3 & 2 & 9 & 2 & 4 & 2 & 9 & 2 & 3 \\
11 & 40 & 15 & 10 & 10 & -6 & -3 & -10 & -22 & -10 & -3 & -6 & 10 & 10 & 15 \\
12 & -13 & 3 & 0 & -10 & 8 & -10 & -10 & 14 & -10 & -10 & 8 & -10 & 0 & 3 \\
22 & 40 & -9 & -10 & 19 & -10 & 2 & 8 & -12 & 8 & 2 & -10 & 19 & -10 & -9.
\end{array}$$

The lower blocks are the transposes. Comparison with the six difference sets shows that $C$ has diagonal 40 and entry $-10$ on every edge of the local graph.

The matrix is positive semidefinite. Exact cyclic Fourier traces and Newton identities give

$$\chi_C(x) = x^{14}(x^2 - 94x + 279)(x^2 - 26x + 135)f(x)^2 g(x)^2,$$

where

$$f(x) = x^6 - 247x^5 + 22466x^4 - 910807x^3 + 15948097x^2 - 111043270x + 189747571,$$
$$g(x) = x^6 - 533x^5 + 105578x^4 - 9736765x^3 + 438572569x^2 - 9062718662x + 68467048091.$$

Every nonzero factor has alternating coefficients. Its value at $-t$ is therefore positive for $t \geq 0$. Since $C$ is real symmetric, it has no negative eigenvalue; the displayed factorization also gives rank 28. An independent rational Schur elimination gives the same twenty-eight positive pivots.

If $v$ is the characteristic vector of a clique of size $c > 0$, then

$$0 \leq v^{\mathsf{T}} C v = 40c - 20\binom{c}{2} = 10c(5 - c).$$

Thus $c \leq 5$. The set $\{A_0, B_6, B_{12}, C_1, C_3\}$ is a five-clique, so the bound is sharp. Moreover, its fourteen translates are linearly independent kernel vectors; nullity fourteen makes them a basis of $\ker C$. Equality in the quadratic-form bound puts every maximum-clique vector in this kernel, and solving the resulting 0–1 constraints in that basis gives precisely the fourteen translates. In particular the seven-clique required by a weight-eight word cannot occur.

### 2.2 The two weight-ten profiles

Let $S$ be a hypothetical weight-ten support, and let $n_i$ be the number of passants meeting $S$ in exactly $i$ points. Code parity makes every occurring $i$ even, and incidence counting gives

$$\sum_i i n_i = 70.$$

At a selected point, pencil parity gives secant degree zero or two. The secant graph induced on $S$ is therefore a union of cycles and isolated vertices. If $m$ is the number of its cycle vertices, then it has $m$ edges, and the other $45 - m$ pairs have passant join. Hence

$$\sum_i \binom{i}{2} n_i = 45 - m.$$

Subtracting half the first identity yields the decisive moment

$$4n_4 + 12n_6 + 24n_8 + \cdots = 10 - m. \tag{1}$$

No $n_i$ with $i \geq 6$ can occur. The degree condition and (1) leave exactly

| $m$ | $(n_2, n_4)$ | shape |
|---|---|---|
| 6 | $(33, 1)$ | four isolated vertices on one four-point passant and six cycle vertices, |
| 10 | $(35, 0)$ | ten cycle vertices; every passant meets $S$ in 0 or 2. |

Indeed two degree-two vertices cannot by themselves form a cycle; in the first case uniqueness of the four-point passant puts all isolated vertices on that line.

The first shape has only four normalized cases. The effective stabilizer of a passant acts as $D_{14}$ on its seven internal points, with four orbits on four-subsets:

| representative | orbit size | $\lvert X\rvert$ | secant degrees in $X$ |
|---|---|---|---|
| 0123 | 7 | 7 | $(3,3,4,4,4,4,4)$ |
| 0124 | 14 | 3 | $(0,0,0)$ |
| 0134 | 7 | 5 | $(2,2,2,4,4)$ |
| 0135 | 7 | 5 | $(2,2,2,3,3)$. |

Here $X$ is the set of internal points passantly joined to all four fixed points; retaining the three unused points of the fixed line only enlarges it. None of the four induced secant graphs contains a six-vertex two-regular subgraph, so the $m = 6$ shape is impossible.

For $m = 10$, fix $P \in S$. The remaining points consist of one point in each of the seven six-point passant fibres through $P$ and an unordered pair among its 35 secant neighbors. The stabilizer $G_P \cong D_{28}$ has thirty-three orbits on the 595 pairs: five of size 7, sixteen of size 14, and twelve of size 28. For one representative of each orbit, add the seven fibre points in order, rejecting a prefix once a passant contains three selected points or a selected point has more than two secant neighbors. The complete certificate has last surviving-depth mass

| depth | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| pair mass | 28 | 35 | 91 | 238 | 203. |

The masses sum to 595, and no representative survives the fifth fibre. Thus $m = 10$ is impossible as well.

Finally, the twelve internal points

$$(1:0:2), (1:3:2), (1:4:5), (1:1:8), (1:4:8), (1:1:7),$$
$$(1:7:12), (1:3:3), (1:9:11), (1:10:11), (1:0:5), (1:8:7)$$

have zero syndrome. Their stabilizer is dihedral of order 24. Therefore $d(K) = 12$.

# 3 Minimum geometry and weighted pairs

Fix one support point $P$. Its seven passant fibres again have six eligible points each, and it has 35 secant neighbors. Pencil parity leaves $s = 0, 2, 4$ secant neighbors. The corresponding multiplicities in the seven passant fibres, followed by the number of secant neighbors, are

$$(5, 1^6; 0), \qquad (3, 3, 1^5; 0), \qquad (3, 1^6; 2), \qquad (1^7; 4).$$

A syndrome-indexed exact check exhausts these four fixed-point domains. Its input sizes are respectively

$$7\binom{6}{5}6^6, \quad \binom{7}{2}\binom{6}{3}^2 6^5, \quad 7\binom{6}{3}6^6\binom{35}{2}, \quad 6^7\binom{35}{2}^2,$$

with disjoint secant pairs in the last domain. The certificate specifies the four partitions, proves their coverage, tests zero syndrome, and deduplicates only after conversion to unordered supports. The four domains contribute $0, 0, 0, 56$ words. The canonical replay generates the four projective $\mathrm{PGL}(2, 13)$-orbits, and intersects each orbit with the supports containing $P$. These four fixed-point slices are disjoint, each has size 14, and their union is checked for equality with the exhaustive 56-word set. The order-28 stabilizer of the fixed point acts transitively on each slice; the stabilizer of a support inside it has order 2. Transitivity, or equivalently the count $56 \cdot 78/12 = 364$, proves global exhaustion.

The replay generates each orbit from its recorded representative and checks every support against $M$. The following constructions identify those orbits without representatives.

### 3.1 One octahedral and three toric families

Parametrize the conic by $(t^2 : t : 1)$, with its usual point at infinity. An internal point $P = (x : y : z)$ defines the fixed-point-free involution

$$j_P(t) = \frac{yt - x}{zt - y}, \qquad J_P = \begin{pmatrix} y & -x \\ z & -y \end{pmatrix}, \qquad J_P^2 = (y^2 - xz)I.$$

The seven secants through $P$ are the seven pairs $\{t, j_P(t)\}$. Thus internal points may be read as perfect matchings of the fourteen conic points.

Fix a chord $e$, carry its endpoints to $\{0, \infty\}$, and let $N_e \cong D_{24}$ be their setwise stabilizer. The three supports over $e$ are

$$S_r(e) = \{(x : 1 : r^{-1}x^{-1}) : x \in \mathbb{F}_{13}^{\times}\}, \qquad r \in \{2, 5, 11\}.$$

Intrinsically, these three parameters satisfy $\chi(r) = -1$ and $\chi(r-1) = 1$. The support $S_r(e)$ is the punctured pencil conic

$$\{y^2 = rxz\}(\mathbb{F}_{13}) \setminus e,$$

so it has size twelve and stabilizer exactly $N_e$. At each displayed point, $\Delta = 1 - r^{-1} = (r-1)/r$ is a nonsquare; hence all twelve points are internal. The set is a codeword for a structural reason. If a passant $AX + BY + CZ = 0$ met $S_r(e)$ with odd multiplicity, the polynomial $Ax^2 + Bx + C/r$ would have a double root. Then

$$B^2 - 4AC = B^2(1 - r)$$

would be a square, contrary to the line being passant. Hence every passant meets $S_r(e)$ in zero or two points.

For the remaining family, take the octahedral subgroup $O \cong S_4$ whose conic-point orbits have sizes six and eight. Let $X_O$ consist of the internal points whose matching has two pairs inside the six-set, three inside the eight-set, and two crossing pairs. In a standard octahedral frame, the six internal-point suborbits have sizes $6, 12, 12, 12, 12, 24$; the displayed matching profile selects one of the twelve-orbits, and direct incidence parity makes it a codeword. Thus $|X_O| = 12$ and $\mathrm{Stab}_G(X_O) = O$.

All four stabilizers have order 24. Each family therefore has $2184/24 = 91 = \binom{14}{2}$ supports, agreeing with the fixed-point exhaustion above.

### 3.2 Exact arity-two reconstruction

For distinct coordinates put

$$c(P, Q) = \#\{S \in \mathcal{H} : \{P, Q\} \subseteq S\}.$$

The exact pair table is

$$\begin{array}{c|cccccc} \rho & 0 & 1 & 3 & 9 & 10 & 12 \\ \hline c & 8 & 6 & 6 & 12 & 7 & 9. \end{array}$$

Only the relations $A_1$ and $A_3$ are fused. For a concurrence-six pair define

$$d_7(P, Q) = \#\{R : c(P, R) = c(R, Q) = 7\}.$$

The $A_{10}^2$ intersection row gives

$$d_7(P, Q) = 2 \quad \text{on } A_1, \qquad d_7(P, Q) = 4 \quad \text{on } A_3.$$

This pair-derived common-neighbor count recovers the full six-color elliptic scheme.

The incidence rows require no coherent closure. Polarity sends $P = (x : y : z)$ to $P^\perp = (-z : 2y : -x)$, and

$$Q \in P^\perp \iff \beta(P,Q) = 0 \iff \rho(P,Q) = 0 \iff c(P,Q) = 8.$$

Consequently the seventy-eight concurrence-eight neighborhoods are exactly the seventy-eight rows of $M$.

The pair data also recovers the code without first naming a row. Let $P_{PQ} = c(P,Q) \bmod 2$ off the diagonal and $P_{PP} = 0$. Then

$$P = A_{10} + A_{12}, \qquad \operatorname{rank} P = 36, \qquad A_0 P = 0,$$

so $\operatorname{im} P = K$. Every coordinate occurs in $364 \cdot 12/78 = 56$ minimum supports, making unary data constant. Pair data suffices and unary data does not; reconstruction arity is exactly two.

# 4 The hidden field, spanning, and automorphisms

For internal points $P = (x_P : y_P : z_P)$ and $Q$, set

$$\Delta(P) = y_P^2 - x_P z_P, \qquad \beta(P,Q) = 2y_P y_Q - x_P z_Q - z_P x_Q,$$

and

$$\rho(P,Q) = \frac{\beta(P,Q)^2}{\Delta(P)\Delta(Q)}.$$

On distinct internal points, $\rho$ takes the six values $0, 1, 3, 9, 10, 12$. Its level relations are the elliptic scheme; write $A_r$ for their adjacency matrices. Polarity identifies $A_0$ with $M$, up to row order.

The needed part of the integral intersection table reduces modulo two to

$$\begin{aligned} A_0^2 &= I + A_9 + A_{10} + A_{12}, \\ A_0 A_r &= 0 \quad (r = 9, 10, 12), \\ A_9^2 &= A_{10}, \qquad A_{10}^2 = A_{12}, \qquad A_{12}^2 = A_9. \end{aligned} \tag{2}$$

These are parity reductions of intersection numbers. The $\mathrm{PGL}(2,13)$-action is transitive on each relation, so fixing one point and substituting one representative of every relation checks all entries.

Put $B = A_9$. Since $A_0 B = 0$, the image of $B$ lies in $K$. If $x \in K$ and $Bx = 0$, then (2) gives

$$0 = A_0^2 x = (I + B + B^2 + B^4)x = x.$$

Thus $B$ is injective on the 36-dimensional space $K$, so $\operatorname{im} B = K$, and $B, B^2, B^4$ all have rank 36.

**Proposition 4.1** (The operator field)**.** *The action of the binary Bose–Mesner algebra on $K$ has image $\mathbb{F}_8$. If $\alpha$ is a root of $t^3 + t^2 + 1$, then*

$$K \cong \mathbb{F}_8^{12}, \qquad (A_9, A_{10}, A_{12}) = (\alpha, \alpha^2, \alpha^4).$$

*The identification of a named $\alpha$ is canonical up to Frobenius.*

Indeed, exact elimination gives $\operatorname{rank}(B+I) = 78$. Restricting the first identity in (2) to $K$ and factoring

$$1 + t + t^2 + t^4 = (t+1)(t^3 + t^2 + 1)$$

therefore gives $B^3 + B^2 + I = 0$. The cubic has no root in $\mathbb{F}_2$, so it is irreducible and makes $K$ a vector space of dimension $36/3 = 12$ over $\mathbb{F}_8$. The squaring identities in (2) give the three displayed scalar operators. This is an operator-field structure on the binary code, not a relabeling as a length-26 code over $\mathbb{F}_8$.

If $N_i$ is the 91-by-78 support matrix of the $i$-th minimum-word orbit, direct pair counting in one intrinsic representative gives

$$(N_1^{\mathsf{T}} N_1, N_2^{\mathsf{T}} N_2, N_3^{\mathsf{T}} N_3, N_4^{\mathsf{T}} N_4) = (A_9, A_9, A_{12}, A_{10}).$$

These are nonzero scalars on $\mathbb{F}_8^{12}$, hence automorphisms of $K$. Thus $\operatorname{im}(N_i^{\mathsf{T}} N_i) = K$ lies in the row space of $N_i$. Its rows already lie in $K$, so every orbit spans $K$.

It remains to determine the scheme automorphisms. Use the four anchors

$$P_0 = (1:0:2), \quad P_1 = (1:1:7), \quad P_2 = (1:0:7), \quad P_3 = (1:1:3).$$

The rigidity mechanism is summarized by

$$\underbrace{(P_0, P_1, P_2)}_{\text{(10,3,9) regular orbit}} \quad \Longrightarrow \quad \underbrace{P_3}_{\text{(3,1,9) is unique}} \quad \Longrightarrow \quad \underbrace{P \longmapsto \big([P = P_i], \rho(P, P_i)\big)_{i=0}^{3}}_{\text{injective on the 78 points}}.$$

The first three satisfy

$$\big(\rho(P_0, P_1), \rho(P_0, P_2), \rho(P_1, P_2)\big) = (10, 3, 9).$$

There are $78 \cdot 14 \cdot 2 = 2184$ ordered triples with this pattern: $A_{10}$ has valency 14 and $p_{3,9}^{10} = 2$. Substitution in the symmetric-square action shows that the stabilizer of $(P_0, P_1, P_2)$ in $\mathrm{PGL}(2, 13)$ is trivial. Since the group also has order 2184, it acts simply transitively on these triples.

For a fixed triple, $P_3$ is the unique point with relation signature $(3, 1, 9)$. The four values

$$\big(\rho(P, P_0), \rho(P, P_1), \rho(P, P_2), \rho(P, P_3)\big),$$

with equality to an anchor recorded on the diagonal, distinguish all 78 points. Consequently a scheme automorphism can be composed with a unique element of $\mathrm{PGL}(2, 13)$ to fix the first three anchors; it then fixes the fourth and every point. Thus the reconstructed scheme has automorphism group $\mathrm{PGL}(2, 13)$.

Every coordinate automorphism of $K$ preserves $\mathcal{H}$. Conversely, $\mathcal{H}$ spans $K$, so every automorphism of $\mathcal{H}$ preserves $K$. The two coordinate-permutation groups coincide.

## 5 Recovery of the ambient conic plane

It remains to recover the plane rather than only its internal-point scheme. Let

$$G = \operatorname{Aut}(\mathcal{H}) \cong \mathrm{PGL}(2, 13), \qquad \Omega = \mathrm{Syl}_{13}(G).$$

A Sylow-13 subgroup has normalizer of order 156, so $|\Omega| = 14$. Conjugation gives a sharply three-transitive action: in the standard model a Sylow subgroup is the unipotent group fixing one point of $\mathbf{P}^1(\mathbb{F}_{13})$, and both $G$ and the set of ordered distinct triples have size 2184. Thus $\Omega$ is intrinsically the recovered conic-point set.

Let $\mathcal{I}$ be the set of 169 involutions of $G$, and put $\Pi = \Omega \sqcup \mathcal{I}$. Index both points and polar lines by $\Pi$. Define incidence by

(i) $U \in U^{\perp}$ for $U \in \Omega$, with no other $\Omega$–$\Omega$ incidences;

(ii) $U \in j^{\perp}$ for $U \in \Omega$, $j \in \mathcal{I}$, exactly when $j$ normalizes $U$;

(iii) $i \in j^{\perp}$ for distinct $i, j \in \mathcal{I}$ exactly when $ij$ is an involution.

These rules are the trace polarity on the adjoint module. Identify the standard plane with $\mathbf{P}(\mathfrak{sl}_2(\mathbb{F}_{13}))$. For traceless $A$, Cayley–Hamilton gives $A^2 = -\det(A)I$. The fourteen singular matrix lines are the nilpotent conic and correspond to the Sylow subgroups; the nonsingular lines correspond bijectively to projective involutions. Moreover

$$\langle A, B\rangle = \operatorname{tr}(AB)$$

is the polar form of the determinant conic. Orthogonality of two nilpotent lines means equality, orthogonality of a nilpotent and an involution means normalization of the Sylow subgroup, and for distinct involutions it is equivalent to their product being an involution. The three intrinsic rules therefore recover $\mathrm{PG}(2, 13)$, its conic $\Omega$, and its polarity.

The 78 original coordinates are also intrinsic inside this plane. A coordinate stabilizer is a nonsplit-torus normalizer $D_{28}$; its unique central involution identifies the coordinate with the involution class of centralizer order 28. The other involution class has size 91. Consequently the old matrix $M$ is exactly the internal–internal block of the recovered polarity matrix. Since Section 3 reconstructed $G$ and $M$ from weighted pairs alone, the whole marked plane is reconstructed at arity two. This completes the proof of Theorem 1.1.

## 6 Proof and verification boundary

The human proof owns the reductions: tangent products produce the local weight-eight graph, line moments produce the two weight-ten shapes, conic pencils produce the toric words, weighted pairs produce the polarity rows, and the adjoint representation produces the ambient plane. Exact execution enters only after those reductions.

The accompanying `verification/` directory contains separate canonical certificates for the theta matrix, moment–stabilizer leaves, weighted-pair reconstruction, toric–octahedral geometry, ambient plane, and hidden operator field. The earlier clique and syndrome programs remain as independent checks but are not theorem-facing proof leaves. The minimum-word exhaustion and four-anchor rigidity retain their original exact replays. The entry point `verify_evidence.py` checks byte counts and SHA-256 hashes before running every program. All arithmetic is exact and deterministic. From the paper root, the public command

```
make check
```

ends with `Paper-IV q=13 evidence:  PASS` and then builds the warning-free PDF. The exact programs are described in `verification/README.md`; the formal package is distributed separately and carries its own README. The Python evidence uses only the standard library, the manuscript build uses a Nix-pinned TEX environment, and the formal package a Nix-pinned Lean environment.

The shared Lean library formalizes the normalized conic and incidence code, the theta inequality, moment compression, pair-neighborhood interface, pencil discriminant identity, and hidden-cubic cancellation. The paper-owned package checks pair-only recovery, unary constancy, the three toric parities, the hidden cubic on the relation-operator image, and both projective-plane uniqueness axioms on the normalized 183-point model. Table 1 records the remaining division of proof.

| claim | proof mode | exact boundary |
|---|---|---|
| dimension 36 | published theorem; Lean theorem | Madison–Wu gives the formula; exact elimination and semantic recovery maps prove rank 42 at $q = 13$. |
| weight 8 | human/classical reduction; exact PSD certificate; Lean implication | Segre's lemma gives the 42-vertex graph. Exact Fourier traces and rational Schur elimination certify the rank-28 form; Lean proves the quadratic-form clique implication. |
| weight 10 | human moment proof; exact stabilizer leaves; Lean algebra | Incidence and pair counts give (1); four $D_{14}$ and thirty-three $D_{28}$ representatives close the two shapes. |
| minimum layer | Lean native evaluation; trusted Python; human transport | Both executions exhaust the 56 fixed-point words; projective transitivity and the double count give all 364. |
| minimum geometry | human conic proof; bounded exact leaf; Lean parity checks | The toric families have a discriminant proof. One octahedral matching profile and the four stabilizers are checked in standard form. |
| pair reconstruction | human polarity proof; trusted exact data; Lean theorem | Pair counts and one common-neighbor count recover the scheme; color-eight neighborhoods recover the rows, and pair parity recovers the code. |
| ambient plane | human adjoint proof; trusted exact replay; Lean incidence gate | The Sylow/involution rules are identified with trace polarity. Lean checks the 183-point plane axioms; the group-theoretic census is exact Python. |
| hidden field and spanning | human algebra; trusted matrix data; Lean theorem | The irreducible cubic makes $K \cong \mathbb{F}_8^{12}$; checked Gram scalars and a kernel-checked image argument prove that every orbit spans. |

Table 1: Proof modes and trust boundaries for Theorem 1.1.

The paper-owned aggregate namespace is `PassantCodeQ13.Gates.Main`; its terminals are

```
incidenceRankAndCodeDimension
fixedPointWeightTwelveExhaustion
minimumOrbitsSpanRhoZeroKernel
ellipticSchemeAutomorphismsAreProjective
pairOnlyReconstruction
toricMinimumSupports
hiddenFieldCubicOnImage
ambientPlaneIncidence.
```

The shared low-weight namespace is `RelativeConicArcs.Gates.PassantCodeQ13`; its terminals are

```
weightEight_semantic_transport
arbitrary_weightTen_profile_transport.
```

Native evaluation compiles and runs finite decision procedures rather than reducing them entirely in the kernel. Under Lean 4.32.0-rc1, the tracked `#print axioms` audit therefore exposes each such use as a declaration-local axiom; the symbolic transport theorems remain kernel-checked relative to those finite leaves. Thus "Lean theorem" in Table 1 means kernel-checked relative to its printed axioms, whereas "Lean native evaluation" and "trusted Python" are executions. Hash checks identify the executed sources and outputs but do not turn either execution into a kernel-checked certificate; no finite leaf in the table is described that way. The formal scope retains three explicit human transports. Projective transitivity and double counting carry the fixed-point exhaustion to all 364 minimum words. Weighted-pair equivariance carries support automorphisms to the polar-relation scheme. The adjoint trace model identifies the abstract Sylow/involution incidence with the conic polarity. Segre's lemma of tangents remains the cited classical input to the weight-eight transport.

## 7 Conclusion

The minimum words retain the geometry that row reduction forgets. Their weighted pairs recover the parity checks and code, then the elliptic scheme, and finally the full conic plane and polarity. The theta form and global line moment explain why this reconstruction first appears in weight twelve; the toric–octahedral families explain what appears there; and the hidden $\mathbb{F}_8$ action explains why each family spans. Thus, in this $q = 13$ instance, the weighted minimum-support 2-section, passant incidence matrix, binary code, elliptic scheme, and marked conic plane are mutually recoverable up to the stated isomorphisms.

Within the Clebsch cubic program, this is the endpoint of the reconstruction theme: a code-theoretic shadow recovers its complete ambient geometry. The argument is nevertheless standalone. It uses neither the earlier papers nor a chosen Clebsch model, and it claims no uniform minimum-distance theorem beyond $q = 13$.

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## AI assistance disclosure


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